SUNY Old Westbury 2026-2027 Undergraduate Catalog

Physics, Minor

Physics is the most fundamental of the sciences, and undergraduate training in physics provides a solid grounding in quantitative problem solving, analytical reasoning, and mathematical modeling. Fundamental knowledge of physics is relevant to a wide variety of majors and careers. A minor in physics is designed for students who have an interest in the fundamental laws of nature, the basic properties of matter, and the nature of space and time.

The physics minor begins with the foundational sequence, Physics 1 and 2, each with their respective labs. Students then take any three additional physics courses, two of which must be upper division (PX 3000 or higher).

Requirements for the Minor in Physics

Students must take at least 17 credits, distributed as follows:

A. Foundational Courses: 8 Credits

For Physics 1 and 2, students may take either the Calculus Track or the Algebra Track. The Calculus Track is recommended for chemistry, math, and computer science majors, as well as any other science majors and pre-health students who have taken or are enrolled in calculus.

Calculus Track

- PX 2200 - Physics 1, Calculus-based Credits: 3
- PX 2201 - Physics 1, Calculus-based lab Credits: 1
- PX 2300 - Physics 2, Calculus-based Credits: 3
- PX 2301 - Physics 2, Calculus-based Lab Credits: 1

Algebra Track

- PX 2000 - Physics 1, Algebra-based Credits: 3
- PX 2001 - Physics 1, Algebra Based Lab Credits: 1
- PX 2100 - Physics 2, Algebra-based Credits: 3
- PX 2101 - Physics 2, Algebra-based Lab Credits: 1

B. Elective Courses: 9 credits

Three additional physics courses, at least two PX 3000 or above.

- PX 2900 - Astronomy Credits: 3 AND
- PX 2901 - Astronomy Laboratory Credits: 1

- PX 3000 - Waves and Modern Physics Credits: 3
- PX 3100 - Mathematical Methods in Physics Credits: 3
- PX 3200 - Computational Physics Credits: 3
- PX 3600 - Mechanics Credits: 3
- PX 3700 - Electromagnetism Credits: 3
- PX 3800 - Relativity Credits: 3
- PX 3900 - Experimental Physics and Electronics Lab Credits: 3
- PX 4100 - Thermodynamics Credits: 3
- PX 4200 - Quantum Mechanics Credits: 3
- PX 5000 - Advanced Topics in Physics Credits: 3